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Entity Relationship Diagram

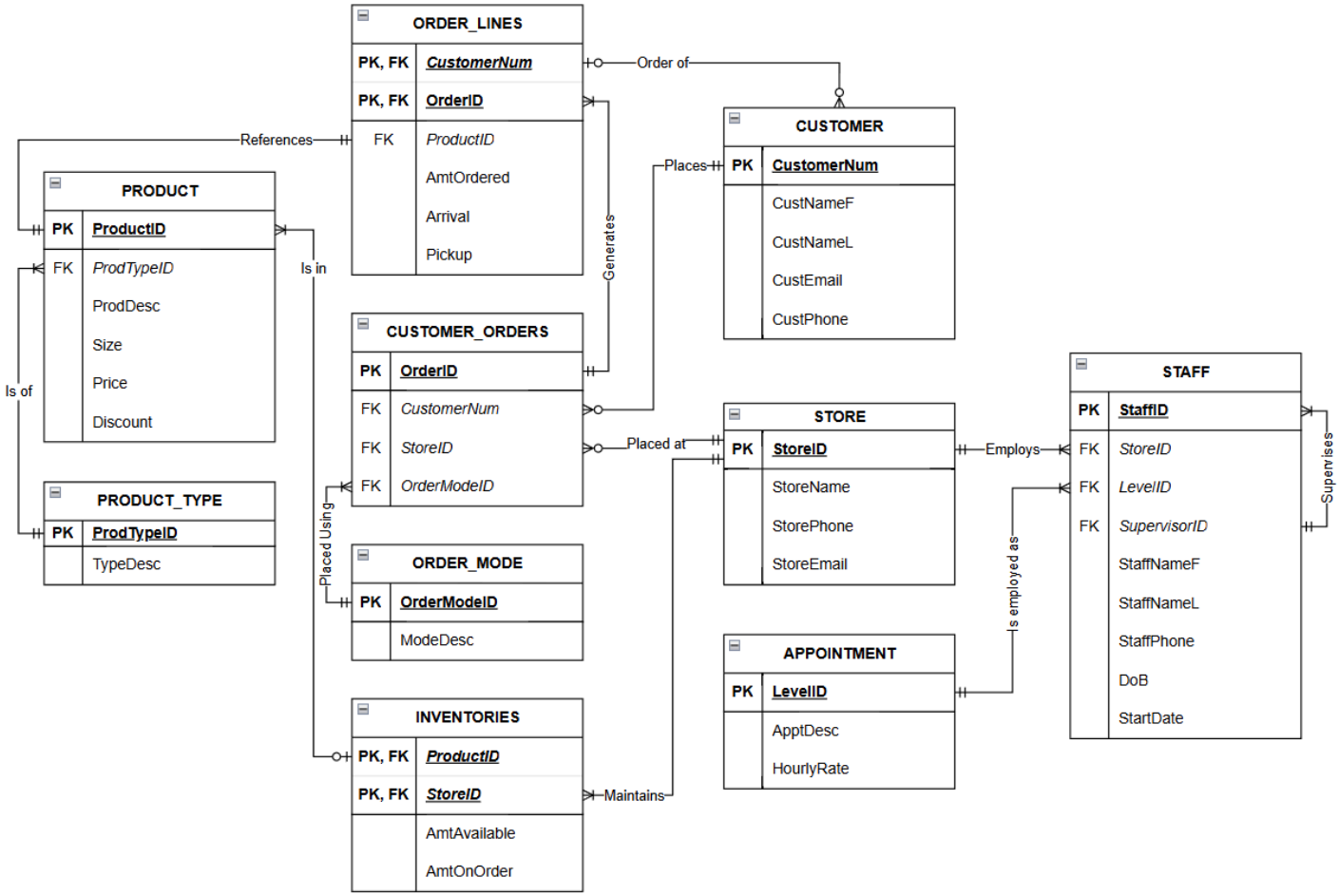

Figure 1.1 Entity Relationship Diagram

Assumptions

We have made the following assumptions regarding the business rules:

- There is a staff member in every appointment level.
- A store will always have a minimum of one inventory.
- There will always be a store.
- Each staff member works at one and only one store.
- Orders can be placed at one and only one store.
- Customers can order a minimum of one thing (thereby generating a minimum of one order line)..
- An order cannot be placed without an order mode, therefore the participation between CUSTOMER_ORDERS and ORDER_MODE is mandatory.
- Product(Size) is in kilograms.
- All staff of an appointment level receive the same hourly rate.

Normalisation

Relation Schema

PRODUCT_TYPE (ProdTypeID, TypeDesc)

ORDER_MODE(OrderModeID, ModeDesc)

APPOINTMENT (LevelID, ApptDesc, HourlyRate)

STORE (StoreID, StoreName, StorePhone, StoreEmail)

CUSTOMER_ORDERS (OrderID,*CustomerNum, StoreID, OrderModeID)*

INVENTORIES (*ProductID, StoreID*, AmtAvailable, AmtOnOrder)

CUSTOMER (CustomerNum, CustNameF, CustNameL, CustEmail, CustPhone)

PRODUCT (ProductID, *ProdTypeID*, ProdDesc, Size, Price, Discount)

ORDER_LINES (*CustomerNum, OrderID*, *ProductID*, AmtOrdered, Arrival, Pickup)

STAFF(StaffID, *StoreID, LevelID, SupervisorID*, StaffNameF, StaffNameL, StaffPhone, DoB, StartDate)

Dependency Diagrams

1. For the PRODUCT_TYPE table, the dependency diagram is:

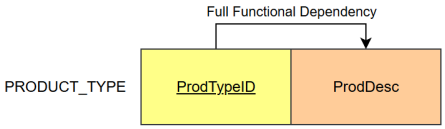

Figure 2.1

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- ProdTypeID → ProdDesk

2. For the ORDER_MODE table, the dependency diagram is:

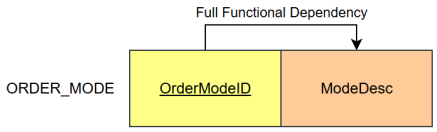

Figure 2.2

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- OrderModeID→ ModeDesc

3. For the APPOINTMENT table, the dependency diagram is:

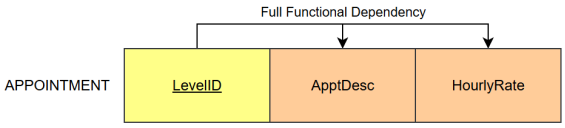

Figure 2.3

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- LevelID→ ApptDesc, HourlyRate

4. For the STORE table, the dependency diagram is:

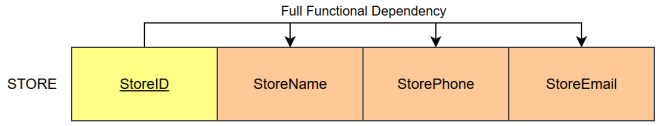

Figure 2.4

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- StoreID→ StoreName, StorePhone, StoreEmail

5. For the CUSTOMER_ORDERS table, the dependency diagram is:

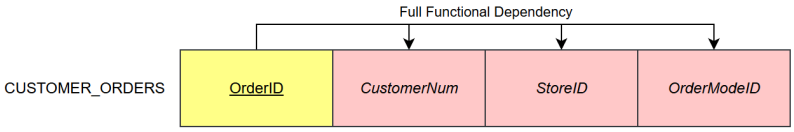

Figure 2.5

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- OrderID→ *CustomerNum, StoreID, OrderModeID*

6. For the INVENTORIES table, the dependency diagram is:

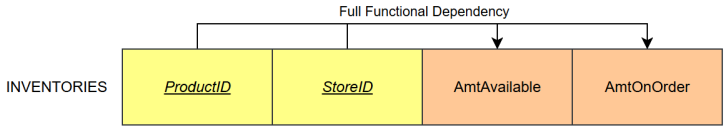

Figure 2.6

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- *{ProductID, StoreID}*→ AmtAvailable, AmtOnOrder

7. For the CUSTOMER table, the dependency diagram is:

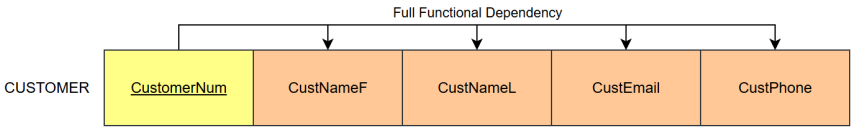

Figure 2.7

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- CustomerNum→ CustNameF, CustNameL, CustEmail, CustPhone

8. For the PRODUCT table, the dependency diagram is:

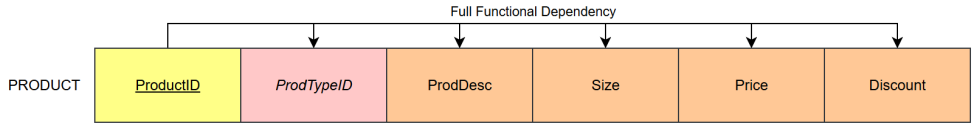

Figure 2.8

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- ProductID→ *ProdTypeID*, ProdDesc, Size, Price, Discount

9. For the ORDER_LINES table, the dependency diagram is:

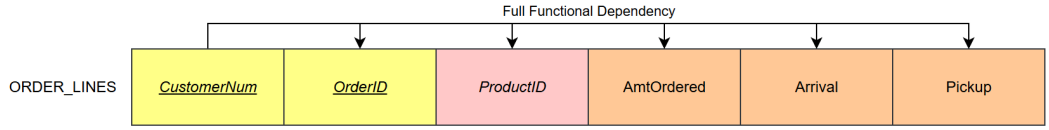

Figure 2.9

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- *{CustomerNum, OrderID}*→ *ProductID*, AmtOrdered, Arrival, Pickup

10. For the STAFF table, the dependency diagram is:

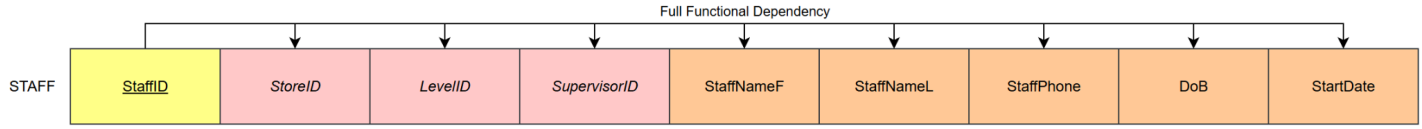

Figure 2.10

The table is in 3NF because it only has full dependency with no partial or transitive dependencies:

- StaffID→ *StoreID, LevelID, SupervisorID*, StaffNameF, StaffNameL, StaffPhone, DoB, StartDate

Relational Database schema

Table Name	Field	Type	Description
STAFF	StaffID	INT(10)	PRIMARY KEY NOT NULL AUTO INCREMENT
	StoreID	INT(10)	FOREIGN KEY REFERENCES STORE(STOREID)
	LevelID	INT(10)	FOREIGN KEY REFERENCES APPOINTMENT(LEVELID)
	SupervisorID	INT(10)	FOREIGN KEY REFERENCES STAFF(STAFFID)
	StaffNameF	VARCHAR(30)	
	StaffNameL	VARCHAR(30)	
	StaffPhone	INT(10)	Format: XXXX-XXX-XXX
	DoB	DATE	Format: DD-MM-YYYY
	StartDate	DATE	Format: DD-MM-YYYY
STORE	StoreID	INT(10)	PRIMARY KEY NOT NULL AUTO INCREMENT
	StoreName	VARCHAR(30)	
	StorePhone	INT(10)	Format: XXXX-XXX-XXX
	StoreEmail	VARCHAR(50)	
APPOINTMENT	LevelID	INT(10)	PRIMARY KEY

			NOT NULL AUTO INCREMENT
	ApptDesc	VARCHAR(500)	
	HourlyRate	FLOAT	
INVENTORIES	ProductID	INT(10)	COMPOSITE KEY REFERENCES PRODUCT(PRODUCTI D)
	StoreID	INT(10)	COMPOSITE KEY REFERENCES STORE(STOREID)
	AmtAvailable	INT(5)	
	AmtOnOrder	INT(5)	
PRODUCT	ProductID	INT(10)	PRIMARY KEY NOT NULL AUTO INCREMENT
	ProdTypeID	INT(10)	FOREIGN KEY REFERENCES PRODUCT TYPE(PRODTYPEID)
	ProdDesc	VARCHAR(500)	
	Size	FLOAT	
	Price	FLOAT	
	Discount	FLOAT(3)	BETWEEN 0 AND 100
PRODUCT_TYPE	ProdTypeID	INT(10)	PRIMARY KEY NOT NULL AUTO INCREMENT
	TypeDesc	VarChar(500)	

CUSTOMER	CustomerNum	INT(10)	PRIMARY KEY NOT NULL AUTO INCREMENT
	CustNameF	VARCHAR(30)	
	CustNameL	VARCHAR(30)	
	CustEmail	VARCHAR(50)	
	CustPhone	INT(10)	FORMAT: XXXX-XXX-XXX
ORDER_LINES	CustomerNum	INT(10)	COMPOSITE KEY REFERENCES CUSTOMER(CUSTOM ERNUM)
	OrderID	INT	COMPOSITE KEY REFERENCES CUSTOMER ORDERS(ORDERID)
	ProductID	INT(10)	FOREIGN KEY REFERENCES PRODUCT(PRODUCTI D)
	AmtOrdered	INT	
	Arrival	DATE	FORMAT: YYYY-MM-DD
	Pickup	DATE	FORMAT: YYYY-MM-DD
CUSTOMER_ORDERS	OrderID	INT	PRIMARY KEY NOT NULL AUTO INCREMENT
	CustomerNum	INT(10)	FOREIGN KEY REFERENCES CUSTOMER(CUSTOM ERNUM)

	StoreID	INT(10)	FOREIGN KEY REFERENCES STORE(STOREID)
	OrderModeID	INT(10)	FOREIGN KEY REFERENCES ORDER MODE(ORDERMODEID)
ORDER_MODE	OrderModeID	INT(10)	PRIMARY KEY NOT NULL AUTO INCREMENT
	ModeDesc	VARCHAR(100)	

Table 1.1